

# SLAM Toolbox: SLAM for the dynamic world

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## Summary

Developments in the field of mobile robotics and autonomous driving have resulted in the use of robots and vehicles in retail stores, hospitals, warehouses, on the roads, and on sidewalks. These deployed areas are both dynamic and frequently massive in scale. The average size of a Walmart store is over 16,000  $m^2$  ([Walmart, 2020](#)) and a single square block in Chicago is over 21,000  $m^2$  ([Heramb, 2007](#)). Retail and warehouse spaces can change drastically throughout the year and the state of roadways can be changing by the hour. Much work has been made to address changing environments in robot perception ([Macenski, Tsai, et al., 2020](#)), but less has been built in open-source to represent maps of dynamic spaces.

For fully autonomous deployed systems to operate in these large and changing environments, they require tools that can be used to accurately map an area specified for their operation, update it over time, and scale to handle mapping of some of the largest indoor and outdoor spaces imaginable. The field of Simultaneous Localization and Mapping (SLAM) aims to solve this problem using a variety of sensor modalities, including: laser scanners, radars, cameras, encoders, gps and IMUs. The most commonly used perception sensor used for localization and mapping in industrial environments is the laser scanner ([Chong et al., 2015](#)). SLAM methods using laser scanners are generally considered the most robust in the SLAM field and can provide accurate positioning in the presence of dynamic obstacles and changing environments ([Cole & Newman, 2006](#)).

Previously existing open-source laser scanner SLAM algorithms available to users in the popular Robot Operating System (ROS) include GMapping, Karto, Cartographer, and Hector. However, few of these can build accurate maps of large spaces on the scale of the average Walmart store. Even fewer can do so in real-time using the mobile processor typically found in mobile robot systems today. The only package that could accomplish the above was Cartographer. However, it was abandoned by Google and it is no longer maintained.

We propose a new fully open-source ROS package, SLAM Toolbox, to solve this problem. SLAM Toolbox builds on the legacy of Open Karto ([Konolige et al., 2010](#)), the open-source library from SRI International, providing not only accurate mapping algorithms, but a variety of other tools and improvements. SLAM Toolbox provides multiple modes of mapping depending on need, synchronous and asynchronous, utilities such as kinematic map merging, a localization mode, multi-session mapping, improved graph optimization, substantially reduced compute time, and prototype lifelong and distributed mapping applications.

This package, `slam_toolbox` is open-source under an LGPLv2.1 at [https://github.com/SteveMacenski/slam\\_toolbox.git](https://github.com/SteveMacenski/slam_toolbox.git) and is available in every current ROS distribution. It was also selected as the new default SLAM vendor in ROS 2, the second generation of robot operating systems, replacing GMapping. SLAM Toolbox was integrated into the new ROS 2 Navigation2 project, providing real-time positioning in dynamic environments for autonomous navigation ([Macenski, Martín, et al., 2020](#)). It has been shown to map spaces as large as 24,000  $m^2$ , or 250,000  $ft^2$ , in real-time by non-expert technicians. An example map can be seen in [Figure 1](#).

# SLAM Toolbox: 动态世界的 SLAM

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## 概括

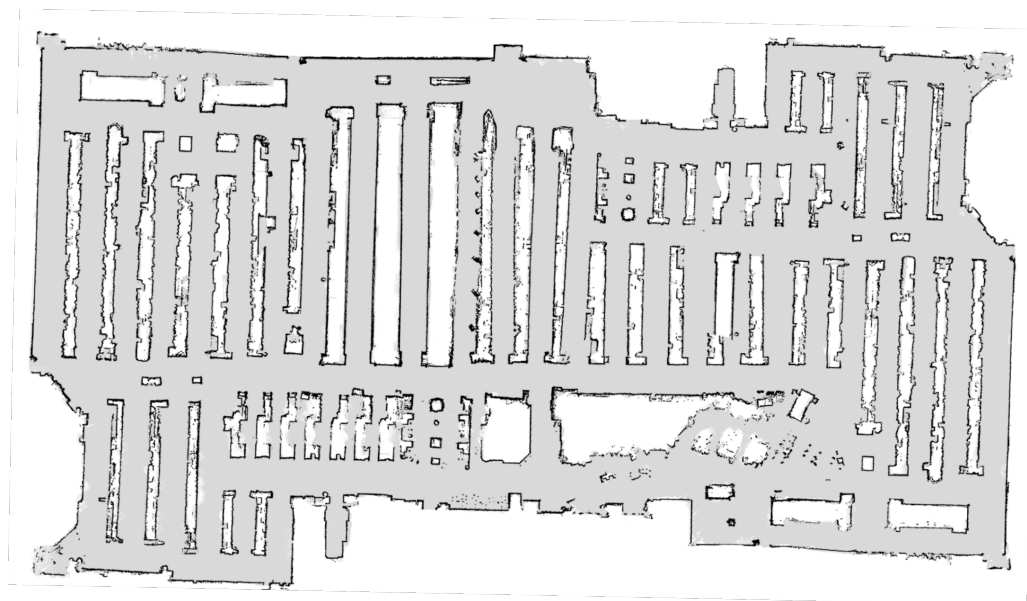
移动机器人和自动驾驶领域的发展导致机器人和车辆在零售商店、医院、仓库、道路和人行道上的使用。这些部署区域都是动态的,而且往往规模巨大。沃尔玛商店的平均规模超过 16,000 家 $m^2$  (沃尔玛, 2020 年), 芝加哥的单个街区超过 21,000 家 $m^2$  (赫兰布, 2007 年)。零售和仓库空间全年可能发生巨大变化,道路状况可能每小时都在变化。为了解决机器人感知中不断变化的环境,已经做了很多工作 (Macenski、Tsai 等人, 2020),但在开源中构建的用于表示动态空间地图的工作却较少。

为了在这些大型且不断变化的环境中运行完全自主部署的系统,它们需要能够准确绘制为其操作指定的区域的工具,随着时间的推移进行更新,并进行缩放以处理一些可以想象的最大的室内和室外空间的绘制。同步定位与建图 (SLAM) 领域旨在使用各种传感器模式来解决这个问题,包括:激光扫描仪、雷达、摄像头、编码器、GPS 和 IMU。在工业环境中用于定位和绘图的最常用感知传感器是激光扫描仪 (Chong 等人, 2015)。使用激光扫描仪的 SLAM 方法通常被认为是 SLAM 领域中最稳健的方法,可以在存在动态障碍物和不断变化的环境的情况下提供准确的定位 (Cole & Newman, 2006)。

之前在流行的机器人操作系统 (ROS) 中可供用户使用的开源激光扫描仪 SLAM 算法包括 GMapping、Karto、Cartographer 和 Hector。然而,其中很少有能够按照沃尔玛普通商店的规模绘制准确的大空间地图。能够使用当今移动机器人系统中常见的移动处理器实时执行此操作的人就更少了。唯一可以完成上述任务的包是 Cartographer。不过,它已被 Google 放弃,不再维护。

我们提出了一个新的完全开源的 ROS 包 SLAM Toolbox 来解决这个问题。SLAM Toolbox 建立在 SRI International 的开源库 Open Karto (Konolige 等人, 2010 年) 的基础上,不仅提供准确的地图算法,还提供各种其他工具和改进。SLAM Toolbox 根据需要提供多种映射模式 (同步和异步),以及运动学地图合并、定位模式、多会话映射、改进的图形优化、大幅减少计算时间以及原型终身和分布式映射应用程序等实用程序。

该软件包 `slam_toolbox` 是在 LGPLv2.1 下开源的,位于 [https://github.com/SteveMacenski/slam\\_toolbox.git](https://github.com/SteveMacenski/slam_toolbox.git),并且在当前的每个 ROS 发行版中都可用。它还被选为第二代机器人操作系统 ROS 2 中新的默认 SLAM 供应商,取代 GMapping。SLAM Toolbox 已集成到新的 ROS 2 Navigation2 项目中,为自主导航提供动态环境中的实时定位 (Macenski、Martín 等人, 2020)。事实证明,非专业技术人员可以实时绘制大至 24,000  $m^2$  或 250,000  $ft^2$  的空间。示例地图如图 1 所示。



**Figure 1:** Retail store map created using SLAM Toolbox ([Macenski, 2019](#)).

## Statement of Need

This software package provides an approach for multi-session mapping and localization. It also utilizes modern graph-optimization techniques and provides first of its kind pose-graph manipulation tools and demo applications. Below we discuss related work and how `slam_toolbox` is meeting a currently unmet need in the community.

## Related Work

SLAM algorithms can be classified into two groups: the earlier algorithms that use the Bayes-based filter approaches ([Thrun et al., 2005](#)), and newer graph-based methods ([Thrun & Montemerlo, 2006](#)). Significant filter-based implementations available as ROS packages are GMapping ([Grisetti et al., 2007](#)) and HectorSLAM ([Kohlbrecher et al., 2011](#)). Cartographer ([Hess et al., 2016](#)) and KartoSLAM ([Konolige et al., 2010](#)) are the major graph-based implementations available.

GMapping is one of the most commonly used SLAM libraries, presented in 2007. It uses a particle filter approach to SLAM for the purpose of building grid maps from 2D lidar data. However, GMapping is not well suited for large spaces and fails to accurately close loops at an industrial scale. Additionally, filter-based approaches cannot be easily reinitialized across multiple sessions.

HectorSLAM relies on lidar scan matching and 3D navigation filter based on EKF state estimation. This method focuses on real-time robot pose estimation and generates 2D map with high update rate. Unlike other mentioned methods, Hector does not use odometry data, which can cause inaccurate pose and map updates when lidar scans arrive at a lower rate, or when mapping large or featureless spaces. HectorSLAM however does not provide loop closure capabilities, making it unsuitable for reliable mapping of large spaces or when using laser scanners with low update rates.

KartoSLAM and Cartographer are both graph-based algorithms that store a graph of robot poses and features. Graph-based algorithms have to maintain only the pose-graph, which

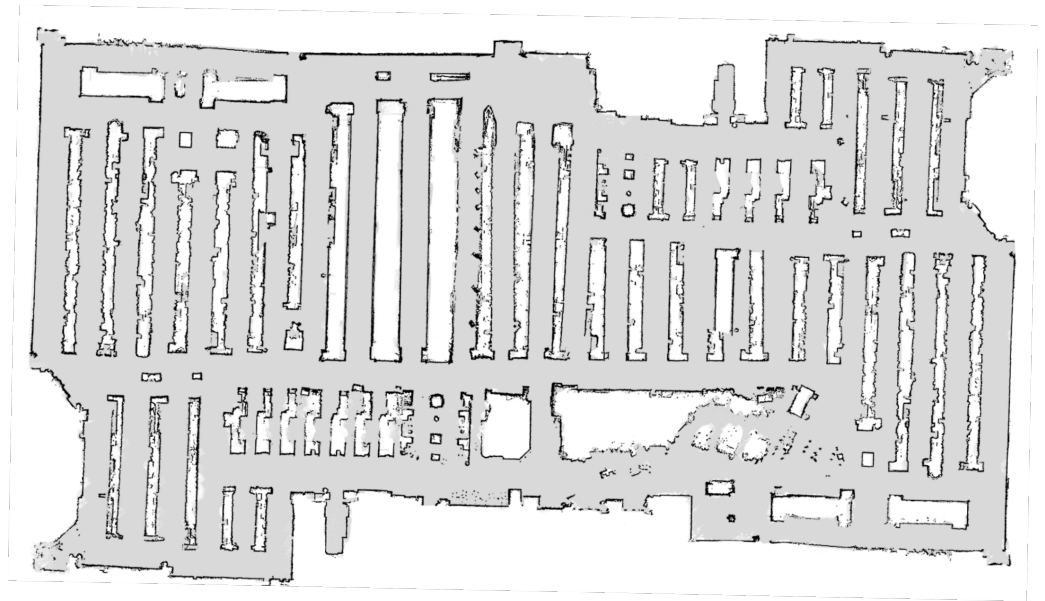


图 1: 使用 SLAM Toolbox 创建的零售店地图 (Macenski, 2019)。

## 需求声明

该软件包提供了一种多会话映射和本地化方法。它还利用现代图形优化技术，并提供首个此类姿势图操作工具和演示应用程序。下面我们讨论相关工作以及 `slam_toolbox` 如何满足社区当前未满足的需求。

## 相关工作

SLAM 算法可以分为两类：使用基于贝叶斯的过滤方法的早期算法 (Thrun 等, 2005) 和较新的基于图的方法 (Thrun & Montemerlo, 2006)。可作为 ROS 包使用的基于过滤器的重要实现是 GMapping (Grisetti 等人, 2007 年) 和 HectorSLAM (Kohlbrecher 等人, 2011 年)。Cartographer (Hess et al., 2016) 和 KartoSLAM (Konolige et al., 2010) 是主要的基于图的实现。

GMapping 是最常用的 SLAM 库之一，于 2007 年推出。它使用 SLAM 粒子过滤器方法，以便从 2D 激光雷达数据构建网格地图。然而，GMapping 不太适合大空间，并且无法在工业规模上准确闭合循环。此外，基于过滤器方法无法轻松地跨多个会话重新初始化。

HectorSLAM 依赖于激光雷达扫描匹配和基于 EKF 状态估计的 3D 导航滤波器。该方法专注于实时机器人位姿估计并生成具有高更新率的二维地图。与其他提到的方法不同，Hector 不使用里程计数据，当激光雷达扫描到达速度较低时，或者在绘制大型或无特征的空间时，这可能会导致姿态和地图更新不准确。然而，HectorSLAM 不提供环路闭合功能，因此不适合大空间的可靠测绘或使用更新率较低的激光扫描仪。

KartoSLAM 和 Cartographer 都是基于图的算法，用于存储机器人姿态和特征的图。基于图的算法必须仅维护位姿图，这

usually makes it efficient in handling resources, especially while building maps of a large scale. KartoSLAM uses Sparse Bundle Adjustment for loop-closure graph optimization. Cartographer consists of front-end, which is in charge of scan matching and building trajectory and submaps, and back-end that does the loop closure procedure. The graph solver used in Cartographer is Google Ceres (Agarwal et al., n.d.). Cartographer provides pure localization mode, when user has a satisfactory map for usage. It also provides data serialization for storing processed submaps only. However, Cartographer has stopped maintenance and support from Google and has been largely abandoned. Further, it fails to build suitable maps for annotation and localization with other localization software packages on robotic platforms without exceptional odometry. The software's unusual complexity makes it challenging to modify or resolve seemingly simple issues, making it not suitable for many applications.

## Features

SLAM Toolbox is able to map spaces effectively using mobile Intel CPUs commonly found on robots well in excess of 100,000  $ft^2$ . It can be done easily using untrained technicians typically hired to deploy robot solutions or remotely using monitoring systems. Some applications have been created to automatically map a space using SLAM Toolbox as well as paired with exploration planners.

It can also serialize a current mapping session and deserialize it at a later time to continue refining or expanding an existing map. This serialization saves the complete raw data and pose-graph rather than submaps, as in Cartographer, allowing a variety of novel tools to be developed and more accurate multi-session mapping. These utilities include manual pose-graph manipulation, whereas a user can manually manipulate the pose-graph nodes and data to rotate a map or assist in a challenging loop closure, shown in Figure 2. It also includes kinematic map merging, the process of merging multiple serialized maps into a composite map. A 3D visualizer plugin was also created to assist in utilization of these tools and the core SLAM library capabilities. Many additional tools and utilities could be developed using this representation as well.

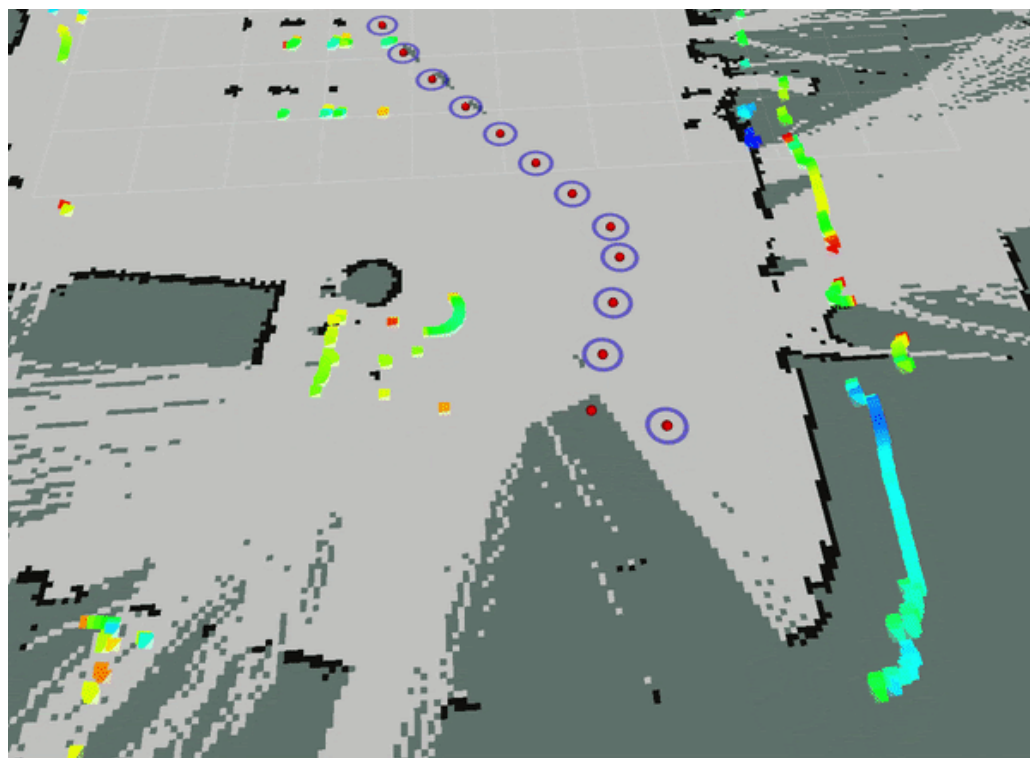


通常使其能够有效地处理资源，尤其是在构建大规模地图时。KartoSLAM 使用稀疏束调整进行闭环图优化。制图器由前端和后端组成，前端负责扫描匹配、构建轨迹和子图，后端则执行回环程序。Cartographer 中使用的图解算器是 Google Ceres (Agarwal 等人, n.d.)。当用户有满意的地图可供使用时，Cartographer 提供纯本地化模式。它还提供数据序列化，仅用于存储已处理的子图。然而，Cartographer 已经停止了 Google 的维护和支持，并且基本上已被放弃。此外，如果没有特殊的里程计，它无法在机器人平台上使用其他本地化软件包构建合适的注释和本地化地图。该软件异常的复杂性使得修改或解决看似简单的问题变得具有挑战性，从而使其不适合许多应用程序。

## 特征

SLAM Toolbox 能够使用超过 100,000 个  $ft^2$  的机器人上常见的移动英特尔 CPU 有效地绘制空间地图。使用通常雇用来部署机器人解决方案或远程使用监控系统的未经培训的技术人员可以轻松完成此操作。一些应用程序已被创建，可以使用 SLAM Toolbox 自动绘制空间地图，并与探索规划器配对。

它还可以序列化当前的映射会话，并在稍后对其进行反序列化，以继续细化或扩展现有映射。这种序列化保存了完整的原始数据和姿态图，而不是像 Cartographer 中那样保存子图，从而允许开发各种新颖的工具和更准确的多会话映射。这些实用程序包括手动位姿图操作，而用户可以手动操作位姿图节点和数据来旋转地图或协助完成具有挑战性的循环闭合，如图 2 所示。它还包括运动学地图合并，即将多个序列化地图合并为复合地图的过程。还创建了 3D 可视化插件来协助利用这些工具和核心 SLAM 库功能。许多其他工具和实用程序也可以使用这种表示来开发。



**Figure 2:** Pose-graph manipulation in progress to manually match a node's laser scan to the map (Macenski, 2019).

It provides 3 major operation modes and executables: synchronous mapping, asynchronous mapping, and pure localization. Synchronous mapping provides the ability to map and localize in a space keeping a buffer of measurements to add to the SLAM problem. This can be advantageous when the quality of the map is of particular importance or when doing offline processing. By contrast, the asynchronous mode will only process new measurements when the last measurement is completed and the new update criteria are met. As a result, this will never lag behind real-time when running complex loop closures. However, the map may not include all valid measurements if processing the last one takes too long. This mode is advantageous when the quality of real-time localization is of particular importance. Both of these modes can be used for multi-session SLAM, the process of reloading a prior session and continuing to refine the pose-graph. Figure 3 shows a map of a large office building created by partially mapping in one session and completing the map in another session. This map has multiple loop closures between the two datasets and was later used with the pure-localization mode to navigate autonomously.

The pure localization mode cannot be used to persist changes in the environment. Instead, it uses a rolling buffer of measurements in the current session and matches them against the original session(s) measurements and pose-graph. The current session's measurements will be added to the pose-graph with new constraints and nodes in the graph. This allows changes in the environment to be embraced to increase localization quality based on new features or moved objects. Over time, the measurements in the rolling buffer will “expire” and be removed from the pose-graph and localization problem, reverting the pose-graph to its original state for that region. The authors refer to this process as elastic pose-graph deformation. An interesting side effect is that the pure-localization mode can be used for effective lidar odometry when paired with no prior mapping session data. It will simply match against its local buffer and keep only a recent view of the environment, allowing lidar odometry to scale to infinite sized spaces.

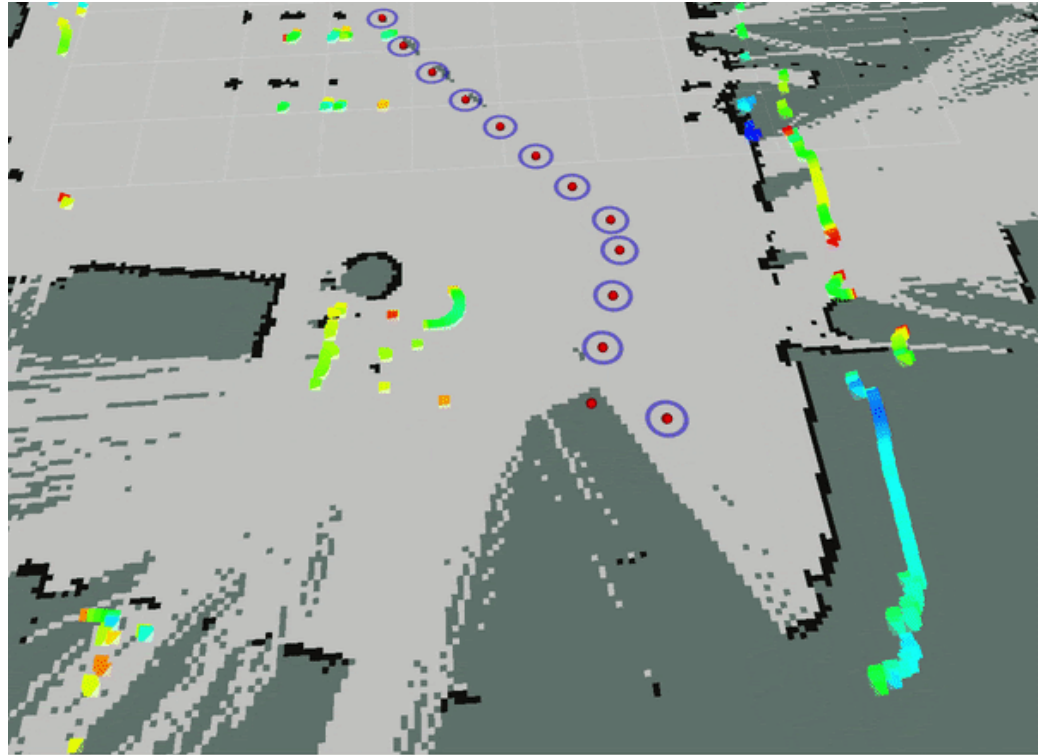


图 2：正在进行位姿图操作，以手动将节点的激光扫描与地图进行匹配（Macenski, 2019）。

它提供了3种主要的操作模式和可执行文件：同步映射、异步映射和纯本地化。同步映射提供了在空间中映射和定位的能力，保留测量缓冲区以增加 SLAM 问题。当地图的质量特别重要或进行离线处理时，这可能是有利的。相比之下，异步模式仅在最后一次测量完成并且满足新的更新标准时才处理新的测量。因此，在运行复杂的循环闭环时，永远不会落后于实时性。但是，如果处理最后一个测量值的时间过长，则地图可能不包含所有有效测量值。当实时定位的质量特别重要时，这种模式是有利的。这两种模式均可用于多会话 SLAM，即重新加载先前会话并继续细化姿态图的过程。图 3 显示了通过在一个会话中部分绘制地图并在另一个会话中完成地图而创建的大型办公楼地图。该地图在两个数据集之间具有多个闭环，后来与纯定位模式一起使用以进行自主导航。

纯本地化模式不能用于持久化环境的变化。相反，它使用当前会话中测量的滚动缓冲区，并将它们与原始会话测量和姿势图进行匹配。当前会话的测量结果将添加到姿态图中，并在图中添加新的约束和节点。这允许接受环境的变化，以根据新功能或移动的对象提高定位质量。随着时间的推移，滚动缓冲区中的测量结果将“过期”并从位姿图和定位问题中删除，从而将位姿图恢复到该区域的原始状态。作者将此过程称为弹性位姿图变形。一个有趣的副作用是，当与先前没有映射会话数据配对时，纯定位模式可用于有效的激光雷达里程计。它将简单地与其本地缓冲区进行匹配，并仅保留最近的环境视图，从而允许激光雷达里程计扩展到无限大小的空间。





**Figure 3:** Large office building map created using multiple SLAM sessions (Macenski, 2019).

Finally, many updates were made to the OpenKarto SLAM library. The measurement matching methods were restructured for a 10x speed-up enabling multi-threaded processing. The provided Sparse Bundle Adjustment optimization interface was replaced with Google Ceres, providing faster and more flexible optimization settings. Additionally, the optimizer interface was turned into a run-time dynamically loaded plugin interface to allow future developers to use the latest and greatest in optimization technologies without modifying the original code. Serialization and deserialization support was enabled to allow for saving and reloading mapping sessions. Finally, new processing modes and K-D tree search were developed to process measurements to enable localization and multi-session mapping. Various smaller improvements and optimizations were also made but excluded for brevity.

## Robots Using slam\_toolbox

SLAM Toolbox has been integrated, tested, and deployed on a number of robot platforms across the world by both industry and researchers. It is also the default SLAM vendor for ROS 2.

A few known examples where SLAM Toolbox has been used or is being used are:

- Simbe Robotics' Tally
- ROBOTIS' Turtlebot3
- Samsung Research America and Russia's research teams
- Rover Robotics' Rover
- Pal Robotics ARI
- Intel's Open Source Group
- Queensland University of Technology researchers



图 3: 使用多个 SLAM 会话创建的大型办公楼地图 (Macenski, 2019)。

最后，对 OpenKarto SLAM 库进行了许多更新。测量匹配方法经过重组，速度提高了 10 倍，支持多线程处理。提供的 Sparse Bundle adjustment 优化界面更换为 Google Ceres，提供更快、更灵活的优化设置。此外，优化器接口被转变为运行时动态加载的插件接口，以允许未来的开发人员在不修改原始代码的情况下使用最新、最好的优化技术。启用序列化和反序列化支持以允许保存和重新加载映射会话。最后，开发了新的处理模式和 K-D 树搜索来处理测量结果，以实现定位和多会话映射。还进行了各种较小的改进和优化，但为简洁起见将其排除在外。

## 使用 `slam_toolbox` 的机器人

SLAM Toolbox 已被行业和研究人員集成、测试并部署在世界各地的多个机器人平台上。它也是 ROS 2 的默认 SLAM 供应商。

SLAM Toolbox 已经使用或正在使用的一些已知示例包括：

- Simbe Robotics 的统计
- ROBOTIS 的 Turtlebot3
- 三星美国研究院和俄罗斯研究团队
- 罗孚机器人公司的罗孚
- 帕尔机器人公司 ARI
- 英特尔开源小组
- 昆士兰科技大学研究人员

- Robosynthesis' EXTRM SC2.0
- Unbounded Robotics UBR-1
- Byte Robotics' Platform
- Unmanned Life's Platform
- MT Robot AG
- Magazino robot datasets
- 6 River Systems

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- Magazino 机器人数据集
- 6 河流系统

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